

Automated minus reference counts on the gold standard: protocol change

frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square-root axes - every asserted gold row

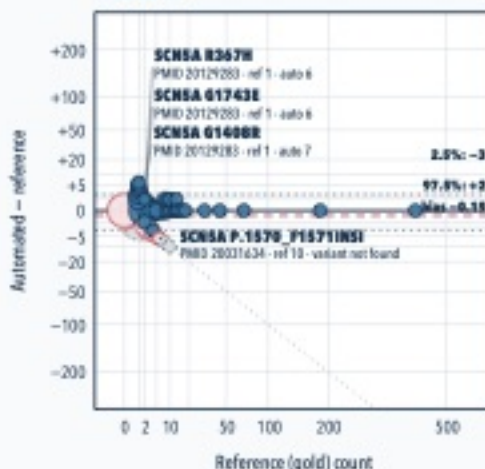
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

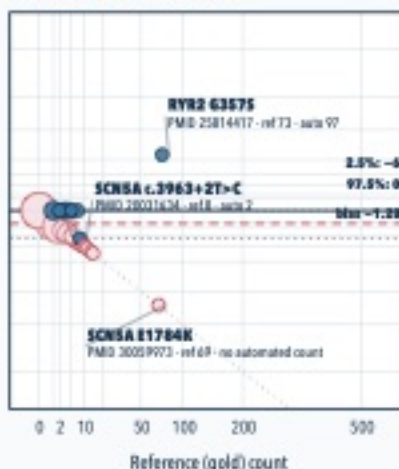
Baseline arm - 20260906_model12_grok43 - locked 2026-09-06 - 12 gene-paper attempts

Carriers



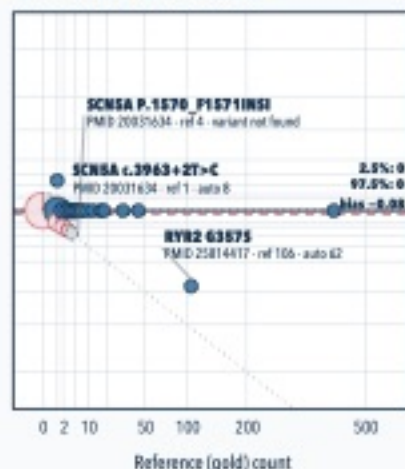
788 gold rows - matched 88.7% - supplied 70.2%
exact 74.5% (zero-baseline 18.5%) - within ± 1 92.5%
bias -0.19 - c2e MAE 0.428 - cond. MAE 0.179 (n 553)
empirical 95% limits -3 to +2 - RMSE 1.08

Affected individuals



788 gold rows - matched 88.7% - supplied 3.0%
exact 28.3% (zero-baseline 17.5%) - within ± 1 81.7%
bias -1.28 - c2e MAE 1.338 - cond. MAE 1.250 (n 24)
empirical 95% limits -6 to 0 - RMSE 3.22

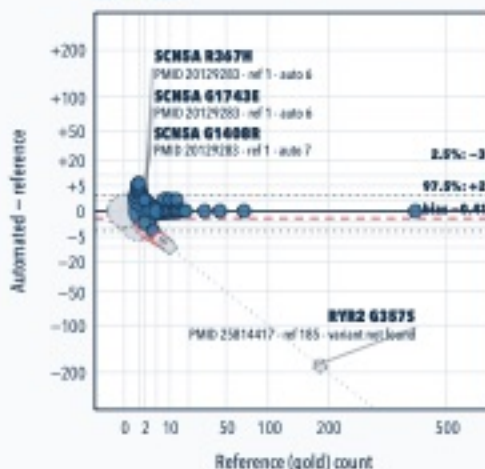
Unaffected individuals



787 gold rows - matched 88.7% - supplied 8.8%
exact 97.6% (zero-baseline 89.1%) - within ± 1 98.9%
bias -0.08 - c2e MAE 0.099 - cond. MAE 0.739 (n 69)
empirical 95% limits 0 to 0 - RMSE 1.61

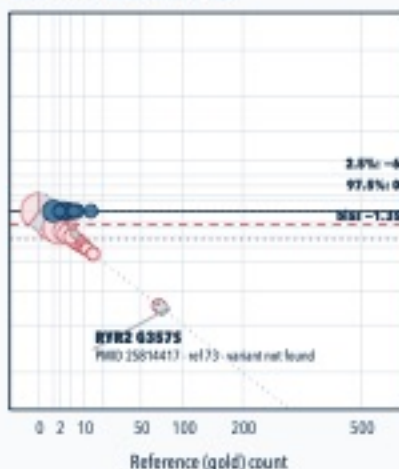
Candidate arm - 20260906_model12_astra_medium_verified - locked 2026-09-06 - 12 gene-paper attempts

Carriers



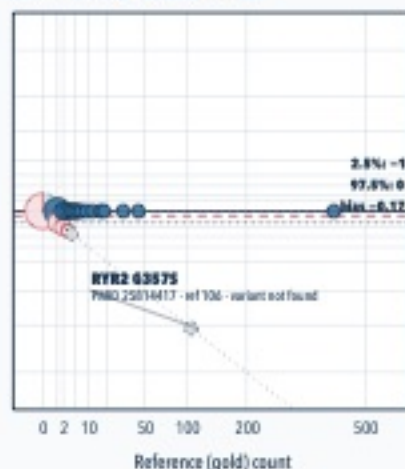
788 gold rows - matched 77.2% - supplied 70.9%
exact 75.4% (zero-baseline 18.5%) - within ± 1 92.8%
bias -0.41 - c2e MAE 0.650 - cond. MAE 0.175 (n 559)
empirical 95% limits -3 to +2 - RMSE 6.68

Affected individuals



788 gold rows - matched 77.2% - supplied 4.7%
exact 22.2% (zero-baseline 17.5%) - within ± 1 82.5%
bias -1.35 - c2e MAE 1.349 - cond. MAE 0.000 (n 37)
empirical 95% limits -6 to 0 - RMSE 4.01

Unaffected individuals



787 gold rows - matched 77.3% - supplied 8.3%
exact 97.3% (zero-baseline 89.1%) - within ± 1 98.9%
bias -0.17 - c2e MAE 0.174 - cond. MAE 0.000 (n 65)
empirical 95% limits -1 to 0 - RMSE 3.79

Candidate - baseline on the same 2363 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 0.428 $\rightarrow$ 0.650 (+0.222)
supplied 553 $\rightarrow$ 559 - exact 587 $\rightarrow$ 594 - bias -0.19 $\rightarrow$ -0.41
rows better 10 - worse 4 - unchanged 774
coverage on matched 79.7% $\rightarrow$ 91.9% (+12.83 pp)

Affected individuals

c2e MAE 1.338 $\rightarrow$ 1.349 (+0.011)
supplied 24 $\rightarrow$ 37 - exact 160 $\rightarrow$ 175 - bias -1.28 $\rightarrow$ -1.35
rows better 18 - worse 5 - unchanged 765
coverage on matched 3.4% $\rightarrow$ 6.1% (+2.65 pp)

Unaffected individuals

c2e MAE 0.099 $\rightarrow$ 0.174 (+0.075)
supplied 69 $\rightarrow$ 65 - exact 768 $\rightarrow$ 766 - bias -0.08 $\rightarrow$ -0.17
rows better 1 - worse 3 - unchanged 783
coverage on matched 9.9% $\rightarrow$ 10.7% (+0.81 pp)

Identity (primary endpoint): recall 88.71% $\rightarrow$ 77.16% (-11.55 pp) - precision 95.10% $\rightarrow$ 96.66% (+1.56 pp) - no registered comparison found; bounds not shown